D.M.A. Pamuditha

Software Engineering Undergraduate

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Summary

Software Engineering student at the National Institute of Business Management (NIBM) seeking opportunities in Software Development, IT Operations and Application Support. Hands-on experience building group projects across web, IoT and machine learning, using Java, Spring Boot, React, JavaScript, Python, PHP and MySQL, including a microservices subscription system, an ESP32 water-quality monitoring platform and a Scikit-learn diagnostic web app. Strong foundation in data structures, algorithms and Git-based collaboration, with the ability to explain design decisions clearly.

Education

BSc (Hons) Computer Science with Software Engineering, *NIBM* 2024 – Present

G.C.E. Advanced Level (Physical Science (Mathematics) Stream), *Prince of Wales' College, Moratuwa*

Skills

Programming — Java | JavaScript | Python | PHP | SQL
Frontend — React | HTML5 | CSS3
Backend — PHP | REST APIs | Microservices
Database — MySQL | Firebase Realtime Database
AI & Machine Learning — Scikit-learn
IoT & Embedded — ESP32 | ESP8266 (NodeMCU) | Arduino IDE | Sensors & Servo Control | I2C
Computer Science — Data Structures (heaps, priority queues, trees, graphs) | Sorting & Graph Algorithms | OOP
Tools & Technologies — Git | GitHub | VS Code | Istio (service mesh basics)
Soft Skills — Team Collaboration | Problem Solving | Technical Communication

Projects

AI Business Autopsy Tool for SMEs Group Final Project
AI-Powered Business Diagnostic Web Application
- Developed the Sales Trend Model (Model 2) using Python, Flask and Scikit-learn as part of a multi-model diagnostic system.
- Built web application features including login, CSV upload, multi-model diagnostic output and PDF export.
- Created a self-contained frontend prototype with CSV parsing for the mid-viva demonstration.

Tech Stack: Python • Flask • Scikit-learn • HTML • CSS • JavaScript

MiniFlix – Microservices Subscription System EAD Coursework
Full-Stack Netflix-Style Subscription and Content Management System
- Developed separate User, Subscription and Content microservices, each with its own database and RESTful APIs.
- Implemented user registration and login with Spring Security (Basic Auth) and subscription plan CRUD operations.
- Built a responsive React admin panel for managing users, content and subscriptions.

Tech Stack: Spring Boot • Spring Security • MySQL • React • Axios • Material UI • React Router • Microservices

ESP32 IoT Water Quality Monitoring System IoT Project
Real-Time IoT Monitoring System with Web Dashboard
- Measured temperature, pH, electrical conductivity and turbidity using an ESP32 and sensor modules.
- Calculated a Water Quality Index (WQI) and a Microplastic Risk Index (MRI) with risk classification.
- Logged data to an SD card in CSV format and synchronized it to Firebase in real time.
- Built a responsive dashboard with 3-second live updates and Chart.js sample comparison.

Tech Stack: ESP32 • Arduino/ESP32 SDK • Firebase • HTML5 • CSS3 • JavaScript • Chart.js

Automated Color Sorting Robotic Arm Group Project
Industrial Manufacturing Prototype – NIBM Robotics Module
- Developed a prototype that detects objects on a conveyor belt with an IR sensor and identifies their color with a TCS3200 sensor.
- Programmed an ESP8266 (NodeMCU) to control a 5-servo robotic arm that sorts items into color-coded boxes.
- Integrated a web dashboard for remote control and real-time counts of each color and total items processed.

Tech Stack: ESP8266 • TCS3200 • IR Sensor • PCA9685 • Servo Motors • Relay • I2C • Web Dashboard

Cycle – Sleep Cycle Optimizer with Nap Planner PDSA Coursework

Browser-Based Sleep and Productivity Planner

- Implemented `productivityForecast.js`, a two-process model (homeostatic pressure and circadian rhythm) that forecasts hourly productivity.
- Designed the night-theme interface in `css/style.css`.
- Used staged Git commits to demonstrate incremental development and defended the design in a viva.

Tech Stack: JavaScript • HTML • CSS • Git

SmartBuild – CI/CD Pipeline Task Scheduler PDSA Coursework

Dependency-Aware, Priority-Based Task Scheduler

- Developed `replanner.py` to re-plan the schedule when a task overruns its estimated duration.
- Collaborated through individual Git branches and pull requests on a scheduler built with graph-based CPM and heap-based task queues.

Tech Stack: Python • Graph Algorithms • CPM • Heaps • GitHub

Zigma Hospital Management System Web Project

PHP-Based Hospital Operations Platform

- Built a PHP and MySQL web application with separate patient and admin interfaces.
- Implemented appointment booking, doctor availability management and news/events management.
- Added security features including CSRF protection, session management and client- and server-side input validation.

Tech Stack: PHP • MySQL • HTML5 • CSS3 • JavaScript

CropCare – AI Plant Disease Detection Presentation Project

AI-Powered Plant Disease Detection for Paddy Farmers

- Presented an AI solution for detecting plant diseases in paddy crops.
- Prepared the technology justification and the spoken presentation script.

Tech Stack: Machine Learning • Presentation

Service Health Monitor and Incident Tracker Personal Project

Python and Flask Operations Monitoring Tool with Live Web Dashboard

- Developed a monitoring tool that checks website and API health (status and response time) and scans application logs for errors.
- Automated incident ticketing with unique IDs and severity levels, creating one ticket per outage and resolving it on recovery.
- Built a live dashboard with auto-refresh, uptime percentage, Chart.js response-time charts, incident filters, Acknowledge/Resolve workflow and light/dark mode.
- Exported SQLite data as JSON and CSV reports for use in Power BI and Excel.

Tech Stack: Python • Flask • SQLite • REST APIs • JSON • HTML • CSS • JavaScript • Chart.js

Certificates

- Docker Training Course for the Absolute Beginner – KodeKloud
- Introduction to Istio (LFS144) – The Linux Foundation
- ITIL 4 Fundamentals – Great Learning Academy
- Developing Secure Software (LFD121) – The Linux Foundation
- Networking Basics – Cisco Networking Academy

Extracurricular Activities

NIBMCodeX 2.0 — *Organizing Committee Member, Finance and Logistics Team* NIBM

School Sports — *Rugby, Athletics and Handball* 2014 – 2020

- **Rugby:** Finished as All-Island Plate runners-up (2015), earned selection to the Sri Lanka under-16 national pool and won individual highlight trophies in yearly tournaments.
- **Athletics:** Won the school sports meet championship for two consecutive years in the 100m, 400m and high jump, and won many zonal and provincial competitions.
- **Handball:** Won the All-Island Cup championship (2015).

Interests

Software Development | IT Operations | Application Support

References

Available on request.